1. We expect the human capital indicators to move with food price inflation. Recent evidence confirms that human capital is a core productive capacity behind agricultural productivity growth in developing economies (e.g. Frontiers in Sustainable Food Systems, 2025)[1]. Productivity is the price mechanism: Fuglie and Wang (2012)[2] show that sustained productivity growth is the main reason real food prices trended downward globally through the twentieth century. Higher human capital stock, therefore, should be associated with lower and more stable food inflation over the medium run.
2. Ideally, we would use the Human Capital Index Plus indicator from the World Bank, along with its dimensions such as education, health and employment. However, the data is mostly missing as this is a newly established indicator. Therefore, we will use proxies for education and health that we collected from the World Bank: life expectancy and tertiary school enrollment.
3. Regression analysis:

- Education has a negative relationship with food price inflation (coef -0.2575). However, the relationship is not significant as the p-value of 0.356 is greater than 0.05.
- Life expectancy has a small positive relationship with food inflation (coef 0.0618). This could be because as people live longer, there is greater demand for food, which drives food prices up. The relationship is not significant as the p-value of 0.850 is greater than 0.05.
- The r-squared of 0.087 is very low, which is expected of social science data.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       BWA_FAO_CP_23014   R-squared:                       0.087
Model:                            OLS   Adj. R-squared:                 -0.009
Method:                 Least Squares   F-statistic:                    0.9099
Date:                Fri, 24 Jul 2026   Prob (F-statistic):              0.419
Time:                        22:02:44   Log-Likelihood:                -64.216
No. Observations:                  22   AIC:                             134.4
Df Residuals:                      19   BIC:                             137.7
Df Model:                           2
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                 7.9974     15.912      0.503      0.621     -25.307      41.302
BWA_education        -0.2575      0.272     -0.946      0.356      -0.827       0.312
BWA_life_expectancy   0.0618      0.323      0.191      0.850      -0.615       0.739
==============================================================================
Omnibus:                        6.359   Durbin-Watson:                   0.946
Prob(Omnibus):                  0.042   Jarque-Bera (JB):                4.232
Skew:                           1.014   Prob(JB):                        0.121
Kurtosis:                       3.712   Cond. No.                         977.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

- Lowess plots suggest a polynomial relationship between food price inflation and life expectancy.

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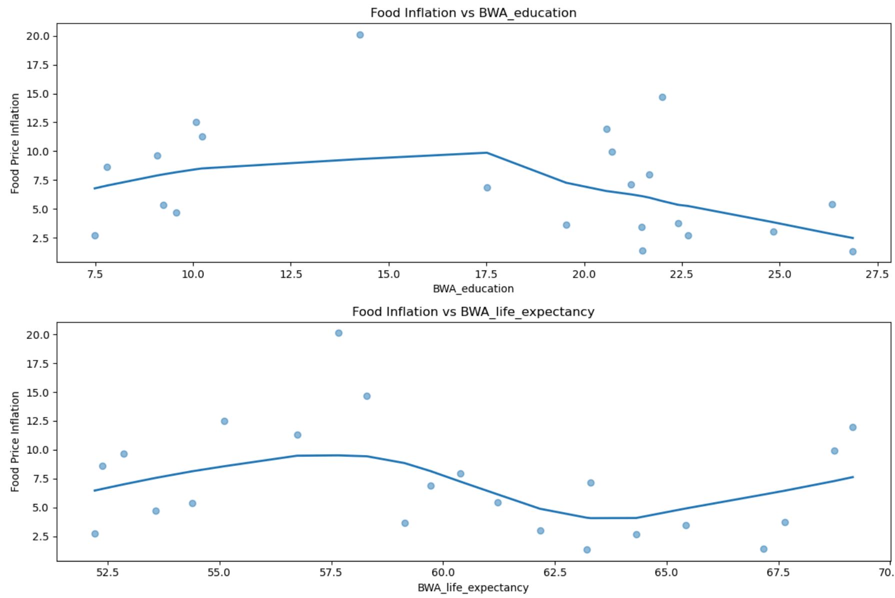
[1] Role of productive capacities on agricultural productivity in BRI countries (2025). Frontiers in Sustainable Food Systems.

[2] Fuglie, K. & Wang, S.L. (2012). Productivity Growth in Global Agriculture Shifting to Developing Countries. Choices,
Quarter 4.



4. Granger Causality Test
- Education (BWA_education) Granger causes food inflation at lag 1 at the 5% significant level.
- Life expectancy does not Granger cause food inflation at any lag, at the 5% significant level.

5. Forward projection
- The forward projection shows sharp decline from 2024, followed by a slow decline of food price inflation. The levels are also similar to those of the food price inflation predicted with Baltic Dry Index, Brent crude oil prices and policy rate, which should give us a little confidence in our results.